

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7 FT 1 HT	(a)	(i)		X marked where the 3 circles intersect		1		1		
		(ii)		Using <u>1</u> station means <u>many</u> possible epicentre locations / using <u>2</u> stations means there are <u>2</u> possible locations for the epicentre location (1) Three stations are needed to eliminate the other options / three circles only cross at one place (1) Accept reference to triangulation		2		2		
	(b)			Station C (1) as it has the smallest circle or diameter or radius / smallest distance or closest [to the epicentre] (1)		2		2		
	(c)			The time delay for B is half that for A or converse (1) Accept time is halved provided one arrival time is correct [Time taken for P wave to arrive at station A = 10 s Time taken for P wave to arrive at station B = 5 s so the time would be] 1:15:15 (1) [The time lag would be half (25 s) the time lag for station A so the time S wave would arrive at station B] = 1:15:40 or $\left[\frac{60}{2} = 30 \text{ s added to } 1:15:10 \text{ to give}\right]$ 1:15:40 (1) N.B. both calculations correct but no statement award 2 marks only		1 1 1	1	3	2	
				Question 7 FT Question 1 HT total	0	7	1	8	2	0